Blood Donation System - Project Report

Abstract

The Blood Donation System is a simple console-based application developed using C++. It is designed to manage donor information efficiently. The system allows users to add donor details, view all registered donors, and search donors based on blood group. This project helps in organizing donor data in a structured way and can be useful for hospitals or blood banks to quickly find suitable donors. The system ensures basic validation like age eligibility and provides a user-friendly menu-driven interface.

Introduction

Blood is a critical resource in medical emergencies, and managing donor information is essential for timely access. Traditional manual systems are time-consuming and prone to errors. This Blood Donation System automates the process of storing and retrieving donor information using C++ and dynamic data structures.

Objectives

1. To develop a system to store donor details efficiently
2. To provide quick access to donor information
3. To enable searching donors by blood group
4. To ensure only eligible donors (age 18–65) are registered
5. To create a simple and user-friendly interface
6. To reduce manual effort in maintaining records

Technologies Used

1. Programming Language: C++
2. Concepts: Object-Oriented Programming, Structures, Classes & Objects, Vectors, I/O Streams
3. Platform: Console-based application

Software Requirements

1. Operating System: Windows / Linux / Mac
2. Compiler: GCC / Turbo C++ / Code::Blocks / VS Code
3. Language Support: C++ (C++11 or above)
4. RAM: Minimum 2 GB
5. Storage: Minimal

Flowchart

START

```
    |
Display Menu
    |
Enter Choice
 /   |   \
Add  View  Search
 |    |      |
Input Display Enter BG
Data  All      |
 |     |       |
Store  Back   Show Results
  \    |      /
      Repeat Menu
         |
        Exit
         |
        END
```

Future Scope

1. Add file handling to store donor data permanently
2. Implement database integration (MySQL)
3. Create a GUI using frameworks like Qt
4. Add update and delete donor features
5. Implement login authentication system
6. Develop a web-based version
7. Add location-based donor search
8. Send SMS/Email notifications